

The prosody of discourse functions: The case of appositive relative clauses in spoken British English¹

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Abstract

This paper aims to explore how discourse functions and prosodic realisations can be related through an analysis of appositive relative clauses (ARCs) in English. Based on a corpus study of spoken British English through the use of the Aix-Marsec and ICE-GB corpora, our aim is twofold: 1) to establish the singularity of ARCs within the global category of parentheticals; and 2) to show that, for one specific syntactic structure, in this instance ARCs, differences in discourse functions correspond not only to differences in morphosyntactic and semantic characteristics as proposed in Loock's (2005, 2007, 2010) threefold discourse typology, but also to differences in prosodic features (regarding tonal, temporal and intensity aspects).

Keywords: *appositive relative clauses, parentheticals, discourse, prosody, spoken English, Aix-marsec, ICE-GB*

1. Introduction

The aim of our research project is to relate discourse functions on the one hand and prosodic realisations on the other hand. In this context, this paper investigates the prosodic characteristics of appositive relative clauses (henceforth ARCs) as a whole, as well as in relation to the different discourse functions they fulfil. Our initial hypothesis, which builds upon Loock's (2005, 2007, 2010) morphosyntactic and pragmatic classification, was that differences in discourse functions might also correspond to differences in prosodic features (tonal, temporal and intensity aspects).

ARCs, also called non-restrictive relative clauses, constitute problematic structures in terms of syntactic description, with analyses ranging from full integration (Jackendoff 1977) to coordination (De Vries 2002) or even orphanage, whether radical (Fabb 1990) or non-radical (Emonds 1979).² From a

discourse point-of-view, Loock (2005, 2007, 2010) shows that ARCs display various functions which can be analysed in terms of discourse (dis)continuity. What this paper investigates is whether prosodic parameters confirm these distinctions.

The potential syntactic non-integration of ARCs has often resulted in an undifferentiated integration into the general category of parentheticals (see Dehé and Kavalova 2007 for an overview of the concept of parentheticals from a syntactic, semantic, pragmatic and prosodic point-of-view). Prototypical prosodic descriptions of parentheticals (which include ARCs) often mention

- surrounding pauses (Armstrong and Ward 1931; Crystal 1969; Cruttenden 1997);
- preceding and following prosodic boundaries (Bolinger 1989; Taglicht 1998);
- lower pitch (Crystal 1969; Bolinger 1989; Local 1992);
- diminished loudness (Crystal 1969; Local 1992);
- higher speech rate (Crystal 1969; Local 1992).

Corpus-based analyses suggest however that parentheticals correspond to a variety of prosodic configurations (e.g. Bolinger 1989), such diversity being possibly related to differences in syntactic, semantic and pragmatic distinctions, as noted for example by Wichmann (2001), Blakemore (2005) for *and*-parentheticals or Dehé (2009) for ARCs, sentential parentheticals and comment clauses. Therefore, the first part of this corpus-based study aims to establish a global comparison of ARCs with parentheticals in general, partially confirming or infirming these previous descriptions (first sub-section in the results).

More centrally, the second part of this study, based on Loock's (2005, 2007, 2010) typology, aims to provide differential analyses depending on the function that the structure fulfils in discourse, indicating the existence of a link between the use in discourse and the prosodic realisation of one syntactic structure. The ARCs in each category were therefore analysed in terms of tonal, temporal and intensity parameters, in search of significant differences in terms of prosodic (non-) integration (second sub-section in the results).

2. Theoretical framework

2.1 Loock's typology of ARCs in discourse

In Loock (2005, 2007, 2010), a classification of ARCs (such as example 1) according to their discourse functions was suggested, providing a semantico-pragmatic positive definition of a syntactic structure generally defined in an

asymmetrical way with Determinative Relative Clauses (DRCs for short, such as example 2), hence other labels like non-restrictive or non-defining, commonly used by grammars, which define ARCs as not being restrictive/defining and not having the characteristics/roles in discourse of DRCs.³

- (1) The students, who are interested in phonology, are also interested in pragmatics-related linguistic phenomena.
- (2) The students who are interested in phonology are also interested in pragmatics-related linguistic phenomena.

In (1), all the students under consideration are interested in phonology and also in pragmatics-related linguistic phenomena, whereas in (2) only those that are interested in phonology are also interested in pragmatics-related linguistic phenomena. What Loock has shown is that ARCs fulfil specific functions that can be defined regardless of the long-established ARC-DRC dichotomy. His research suggests that three main categories can be distinguished: continuative ARCs, subjectivity ARCs and relevance ARCs. Below, we provide a definition of each category along with examples.

2.1.1 *Relevance ARCs*

As is exemplified in the examples given in (3),⁴ relevance ARCs correspond by definition to the pragmatic constraint that a speaker has when s/he needs to convey information that may be known by the addressees. Though the antecedent has been identified from a purely referential point of view, it is not sufficiently determined to be used alone in discourse, at least for some of the targeted addressees (following Prince's (1981) and (1992) taxonomy of given and new information, extended by Loock (2005, 2010), the information has an indeterminable information status). The use of an ARC aims to ensure the relevance of the antecedent and/or the subject-predicate relation within the main clause (henceforth MC).

- (3) a. he was convinced # the battle # for the hearts # and minds of the people # was being won # especially # among the Ovambo # who form the majority # of SWAPO's support (AM_A0905G)
- b. Well I got involved through Celeste who <,> I used to dance with with London Contemporary <,> going back <,> twenty years or so <,> (ICEGB_s1a-002 004)

2.1.2 *Subjectivity ARCs*

An ARC is also used by speakers to convey information that represents a comment, a judgment or an assessment on the informational content of the MC (nominal or sentential antecedent), either by themselves or by somebody else.

This information, which does not belong to the informational foreground (“at-issue content”, see Potts 2005) and is not conveyed by an extra independent clause, is backgrounded (“not at-issue content”, see Potts 2005) through the use of an ARC and therefore represents a rupture with the discourse topic. (4) shows two examples of subjectivity ARCs:

- (4) a. Israelis # have sympathy and liking for Americans # which is just as well # since the country is swarming # with transatlantic visitors (AM_A0704G)
- b. And I uh used to get maybe it was five five pounds as a birthday present which was always good to have because it was real <unclear-words> cash you know (ICEGB_s1a-076 125)

2.1.3 *Continuative ARCs*

Continuative ARCs had already been defined by Jespersen (1970) and Cornilescu (1981) among others, but definitions are not systematically interchangeable as they mix syntactic and discursive considerations, many authors applying the label ‘continuative’ to describe relative clauses with sentential, non-nominal antecedents. Loock’s definition is purely discursive: a continuative ARC is defined as the syntactic vehicle used to narrate an event that is successive to a first event depicted in the MC. Continuative ARCs therefore “make narrative time move forward”, possibly establishing a causality link. This is quite an unusual function for an embedded/subordinate clause (Depraetere 1996), narrative dynamism generally being restricted to independent clauses; our position is more in line with Thompson (1987), who shows that temporal ordering and independent clauses do not systematically go hand in hand. Examples in (5) illustrate this category:

- (5) a. Northern Scotland will have occasional light rain which will be followed during the day by colder but still mainly cloudy weather # with a few sleet and snow showers (AM_M0801G)
- b. You can understand such a lot of poetry at a child’s level which later on becomes much more complicated (ICEGB_s1b-048 039)

This category is very specific and clearly in opposition with the two other categories as far as the hierarchisation of the informational contents is concerned: both the MC and the ARC belong to the informational foreground, a phenomenon which has paved the way for the suggestion found in the literature that ARCs share syntactic characteristics with independent clauses. Some linguists indeed consider that ARCs are integrated at discourse level only and do not belong to the syntax of the sentence (e.g. Ross 1967; Emonds 1979; McCawley 1982 or Fabb 1990 among others).

2.2 Prosodic framework

The prosodic analyses carried out in this study are in line with a conception of prosody as a complex system not to be reduced to tonal considerations alone (for examples of such complex conceptions, see Couper-Kuhlen and Selting 1996; Ladd 1996; Di Cristo and Hirst 1998; Di Cristo 2000). We favour a holistic view of prosody which aims to take into account not only those aspects related to tone and intonation, but also other elements related to temporal phenomena, intensity and voice quality (see Du Bois et al. 1993; Campbell and Mokhtari 2003; Gobl and Ní Chasaide 2003 or Campbell 2004 regarding this last concept).

This study is also concerned with the distinctions between levels of representation as presented in Hirst et al. 2000). The phonetic analyses carried out in the present study, and more particularly those concerning tonal aspects, rely on the key distinction between the acoustic and phonetic levels. In this perspective, a preliminary phase to tonal analysis involved MOMEL modelling (*cf.* Hirst and Espesser 1993 and Hirst et al. 2000), which aims at factoring out any micro-segmental characteristics (the “micro-prosodic component” described in Di Cristo and Hirst 1986). The resulting curve (see Figure 1) is thus similar to that found on a sequence of vowels and sonorant segments (known to have the smallest micro-melodic effect) and constitutes the “macro-prosodic component”, a phonetic construct as opposed to the purely acoustic F0 curve. For instance, such modelling neutralises the lowering effects of voiced obstruents, or the pitch skip observed on vowel onsets preceded by a voiceless consonant (see Di Cristo 1978 for a more detailed analysis of micro-melodic effects).

Our analyses also tackle the question of “orthogonal dimensions” (Ladd 1996). More specifically, the concepts of pitch range and intensity range were taken into account. This concept of range can be divided into two parameters, level and span, illustrated in Figure 2.

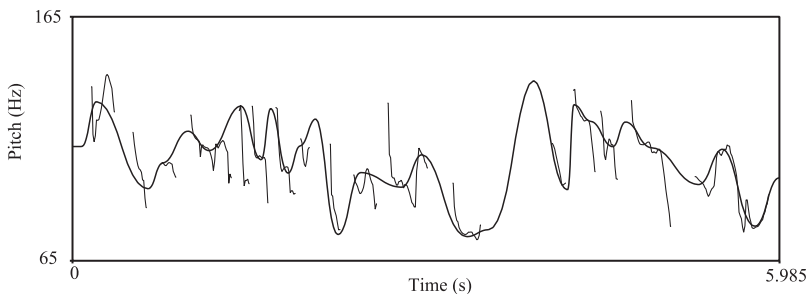


Figure 1. MOMEL modelling of the F0 curve

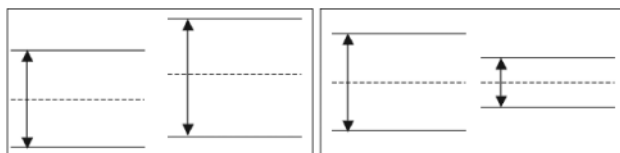


Figure 2. *Difference in levels (left box) and spans (right box)*

For instance, in the tonal system (pitch range), the traditional linear succession of high and low tones can be analysed as taking place within a global frame, or space. Pitch level variations (Ladd 1996 or “pitch register” in Cruttenden 1997) are related to global “raising or lowering of the contour in the F0 space” (Gussenhoven 2004: 76). Pitch span (Ladd 1996 or “key” in Cruttenden 1997) is related to the distance between the highest peaks and the lowest troughs in the F0 contour. Intensity range was treated in an analogous way, with intensity level corresponding to a global raising or lowering of the contour in the intensity space, and intensity span to the distance between the highest peaks and the lowest troughs in the global intensity contour.

Modifications in pitch level and span are traditionally associated with parentheticals (prosodic compression; *cf.* Wichmann 2000) and are thus to be taken into account in the prosodic characterisation of ARCs. Intensity is described as following similar patterns and therefore received particular attention within this study (in spite of difficulties inherent to speaker-to-microphone distance variations, which, though non-linguistic in nature, do impact global intensity measurements).

3. Data and methodology

3.1 Data

This study relies on the use of two spoken British English corpora, providing extensive access to a wide variety of speech types: (i) the Aix-MARSEC corpus (*cf.* Auran, Bouzon and Hirst 2004) and (ii) the ICE-GB corpus (*cf.* Greenbaum 1996). The Aix-MARSEC consists of BBC recordings from the 1980s and provides natural sounding but mostly scripted English by 53 speakers. It constitutes a second evolution from the original SEC (Spoken English corpus, *cf.* Knowles, Wichmann and Alderson 1996), the MARSEC (Machine Readable SEC) constituting the first one (Roach et al. 1993). The ICE-GB corpus contains 1,000,000 words of written and spoken English from the 1990s, the latter being both scripted and unscripted. In this study, we focused on the unscripted spoken parts of the corpus, *i.e.* 250 texts (out of the 300 in the

Table 1. *Analysed and original corpora*

Corpus	Relevance ARCs	Subjectivity ARCs	Continuative ARCs	Non-congruent labels	Total
ICE-GB	63 (79)	37 (67)	9 (17)	0 (5)	109 (168)
Aix-Marsec	39 (56)	8 (10)	1 (1)	0 (8)	48 (75)
Total	102 (135)	45 (77)	10 (18)	0 (13)	157 (243)

Table 2. *Distribution of scripted and unscripted ARCs in the analysed corpus*

	Relevance ARCs	Subjectivity ARCs	Continuative ARCs	Non-congruent labels
Scripted	69	15	4	88
Unscripted	33	30	6	69
Total	102	45	10	157

spoken section) for a total of approximately 500,000 words (source: <http://www.ucl.ac.uk/english-usage/projects/ice-gb/design.htm>).

From the Aix-MARSEC and ICE-GB corpora were extracted a total of 243 ARCs, which were manually labelled by the authors in relation to their discourse function (relevance, subjectivity or continuative ARCs), and the examples for which labels were non-congruent (for a total of 13; see table 1) were excluded from the analyses. The method of extraction used involved an automatic search for relative *wh*-pronouns and *that* in both corpora; Ø (zero pronoun) was therefore excluded, in line with reference grammars, which consider that ARCs are never introduced by a zero pronoun (Quirk et al. 1985; Huddleston and Pullum 2002). Based on Loock (2005), the distinction between ARCs and DRCs was based on the referential (in)dependence of the antecedent (pragmatic criterion): an antecedent that has found its reference independently of the embedding of the relative clause therefore triggered a classification of the relative clause as an ARC (also see Loock 2010 for more detail). Where needed, prosodic phenomena (pitch break and/or pause) were also taken into account to complement this classification.

For technical reasons, not all examples could undergo prosodic analyses, because the quality of the sound file was not good enough for an acoustic analysis (background noise, bad recording conditions) or because of overlapping voices, laughter, mumbling and/or performance errors. Out of the initial 243 occurrences, only 157 could undergo an acoustic analysis and could be used for our study. Table 1 below provides the exact figures for the analysed corpus and the original corpus (in between brackets). Table 2 presents the distribution of scripted and unscripted ARCs in the analysed corpus.

3.2 Methodology

Our analyses involved a two-step procedure, with a discourse annotation followed by a prosodic annotation.

3.2.1 Discourse annotation

The discourse annotation of the corpus covered 5 discourse parameters: the extracted ARCs were manually annotated according to

- ARC type;
- position in the sentence (initial,⁵ middle, or final);
- information status of the antecedent and information status of the ARC (based on Prince's 1981 and 1992 taxonomy of given/new information (discourse new/old, hearer new/old, inferrables), extended in Loock 2005 to include indeterminables);
- nominal/sentential status of the antecedent.

While our global project aims to take the entire set of parameters into account, this study focuses on ARC type and ARC position in the sentence, the latter potentially influencing the prosodic realisation of the following unit (by definition, final ARCs precede new sentences, which is not the case for middle ARCs). Table 3 presents the distribution of middle and final ARCs in the analysed corpus. Although prosodic features such as pauses or pitch-breaks were sometimes used to tease apart DRCs from ARCs, only the transcripts were used for the discourse annotation.

3.2.2 Prosodic annotation

The relevant recordings from both corpora were semi-automatically analysed, using original scripts in *Praat* (Boersma 2001; Boersma and Weenink 2006) and covering up to 50 prosodic parameters in total concerning the tonal, temporal and intensity systems.

The prosodic annotation implied the segmentation of relevant parts of the speech signal into intonation units (henceforth IUs). This segmentation was carried out manually by the first author, and resorted to phonetic characteristics classically found in discourse studies based on authentic speech: a coherent

Table 3. *Distribution of middle and final ARCs in the analysed corpus*

	Relevance ARCs	Subjectivity ARCs	Continuative ARCs	Non-congruent labels
Middle	31	9	0	40
Final	71	36	10	117
Total	102	45	10	157

(i.e. continuous) intonation contour, the presence of a nuclear accent and optionally pauses, onset pitch reset and final lengthening (Chafe 1987; Du Bois et al. 1993; Wightman et al. 1992; Cruttenden 1997; Gussenhoven 2004). Based on these phonetic criteria, our intonation units do not differ significantly from Intonational Phrases (henceforth IPs; Nespor and Vogel 1986; Beckman et al. 2005).

F0 threshold detection and modelling with the MOMEL algorithm was carried out automatically using a dedicated Praat plugin (Hirst 2007). Threshold values and MOMEL targets were then manually checked and corrected where necessary. Eventually, the selection of the stable part of the nucleus of the first accented syllable (onset) and of the last syllable (offset) was done manually, while the extraction and computation of all parameters was automatic.

For each ARC, and for each preceding and following IU, the following measurements were taken:

- Tonal system (values in hertz and semitones):
 - mean F0 (pitch level)
 - minimum F0
 - maximum F0
 - F0 standard deviation (pitch span)
 - F0 on the first accented syllable (onset)
 - F0 on the last syllable (offset)
 - difference between previous IU offset and ARC onset (differential onset)
 - difference between ARC offset and next IU onset (differential offset)
- Temporal system:
 - duration (raw and z-standardised)
 - difference between previous IU z-standardised duration and ARC z-standardised duration
 - difference between ARC z-standardised duration and next IU z-standardised duration
 - silence duration before ARC
 - silence duration after ARC
- Intensity system:
 - global intensity mean (intensity level)
 - global intensity standard deviation (intensity span)

All 157 ARCs in our corpus were realised as one or more IUs, in agreement with the trend noted in Dehé (2009), whereby longer parentheticals (sentential parentheticals and ARCs) strongly tend to be mapped onto separate intonation units (IPs in Dehé 2009). Whenever an ARC spanned several IUs (such as in examples 3b, 4a and 5a), onset measurements were carried out on the onset of

the first IU, and offset measurements on the offset of the last IU; all other measurements were globalised over the ARC.

Along with their absolute values in Hertz, all F0 measurements were normalised using a semitone (henceforth ST) scale centred on the average pitch value for each speaker. For instance, for a given speaker, a pitch level value corresponding to the average pitch measured on the relevant recordings thus provided a normalised value of 0 ST; a pitch level value double the average pitch measured on the relevant recordings provided a normalised value of 12 ST.

For want of phoneme-level alignment for the ICE-GB corpus, phone durations, speech rates and speech rate variations could not, in this study, be computed for the whole corpus. Consequently, only the raw temporal parameters mentioned above, with a total of 46 prosodic parameters, were retained, thus leading to a total of 7,222 observations. Among those 46 parameters, the analyses in this paper focused on the more classical parameters pertaining to the description of parentheticals and topical structure (pitch level, pitch span, intensity level, intensity span and onset values).

Formal statistical analyses were eventually carried out within the R environment and software.

4. Results

After briefly presenting a global prosodic description of the ARCs in our corpus, this section exposes the differential results depending on the ARCs' discourse functions.

4.1 *Global prosodic description of ARCs*

As mentioned above, we restricted our analyses to some of the most frequently mentioned prosodic characteristics of parentheticals:

- pitch level and span;
- onset value: both in absolute (normalised F0 value) and relative terms (normalised differential between ARC onset and offset of the preceding IU);
- intensity level and span.

The normalised pitch level in ARCs is significantly lower (−0.6 semi-tones (ST)) than in preceding (0.3 ST; Kolmogorov-Smirnov Test:⁶ $D = 0.3758$, $p < 0.001^{**}$) and following (−0.07 ST; KS Test: $D = 0.242$, $p < 0.001^{**}$) intonation units (IUs). However, the pitch span of ARCs (10.71 ST) is not significantly different from that of the preceding (10.44 ST; KS: $D = 0.0828$, $p \approx 0.655$) or following (10.57 ST; KS: $D = 0.0547$, $p \approx 0.978$) IUs.

Table 4. ARC pitch, intensity and onset values compared with preceding and following IUs

Prosodic features		Preceding IU	ARCs	Following IU
Pitch	Level	0.3 ST (KS: $D = 0.3758$, $p < 0.001$)	-0.6 ST	-0.07 ST (KS: $D = 0.242$, $p < 0.001$)
	Span	10.44 ST	10.71 ST	10.57 ST
Intensity	Level	68.29 dB		68.29 dB
	Span	8.095 dB		8.134 dB
Onset	Absolute		1.267 ST	
	Differential		1.539 ST	

In addition, we noticed no significant difference in intensity levels between the ARCs (67.04 dB) and their preceding (68.29 dB; KS Test: $D = 0.1083$, $p \approx 0.316$) and following (68.29 dB; KS Test: $D = 0.0892$, $p \approx 0.56$) IUs. Intensity span results, similarly, do not indicate any significant differences between the ARCs (8.142 dB) and their preceding (8.095 dB; KS Test: $D = 0.121$, $p \approx 0.2$) and following (8.134 dB; KS Test: $D = 0.0764$, $p \approx 0.749$) IUs.

It is interesting to note that the onset (first accented syllable) absolute value is clearly positive (1.267 ST). The onset differential (difference between ARC onset and the last syllable of the preceding IU) also displays a positive value (mean = 1.539 ST). Taken together, these results indicate that ARC onset F0 values are globally above the speakers' average pitch levels, while the offset values of the preceding IUs are only very slightly above (with a mean of 0.3 ST). These values traditionally associated with discourse discontinuity (e.g. Wichmann 2000) will be further commented upon in the discussion. Table 4 summarises these results.

ARC position within the sentence (middle vs. final in our corpus) has no significant influence on pitch level (KS: $D = 0.1026$, $p = 0.877$), pitch span (KS: $D = 0.1113$, $p = 0.811$), intensity level (KS: $D = 0.1707$, $p = 0.312$) nor intensity span (KS: $D = 0.167$, $p \approx 0.309$). Similarly, onset absolute (KS: $D = 0.2058$, $p \approx 0.139$) and differential (KS: $D = 0.2402$, $p \approx 0.054$) values are not influenced by ARC position.

These results, with lower pitch levels and increased onset values for ARCs as a whole, do not reflect the variety of realisations that can be found if we study ARCs differentially, that is by discriminating between the different discourse categories as defined in Loock's taxonomy. We therefore now turn to these differential results.

4.2 Differential results

The parameters mentioned in our global results (pitch level, pitch span, intensity level, intensity span and onset values) were systematically taken into

Table 5. *Normalised pitch level by ARC type and differences with preceding and following IUs*

ARC type	Preceding IU	ARC type	Following IU
Continuative	-0.259 ST (KS: $D = 0.6$, $p \approx 0.052$)	-0.929 ST	0.128 ST (KS: $D = 0.5$, $p \approx 0.168$)
Relevance	0.447 ST (KS: $D = 0.4118$, $p < 0.001$)	-0.609 ST	-0.045 ST (KS: $D = 0.2647$, $p < 0.005$)
Subjectivity	0.099 ST (KS: $D = 0.2889$, $p < 0.05$)	-0.529 ST	-0.157 ST (KS: $D = 0.3111$, $p \approx 0.05$)

account. Given that no category displays any significant difference with its surrounding IUs in terms of pitch span, no differences could be found between ARC types regarding this feature. This result is consistent with the global results given above. Still in line with our global results, our distinctive analyses reveal no difference in intensity level or span with surrounding IUs. The rest of this section will be devoted to those parameters for which differences between ARC types were evidenced.

As far as differences between ARC types are concerned, the first interesting result concerns pitch level. Contrary to the global trend mentioned above (with lower global values for ARCs), pitch level for continuative ARCs is not significantly lower than that of preceding and following IUs (see table 5).

Noteworthy results were provided by the analysis of onset values. As can be seen on figure 3, higher normalised mean onset values were found for subjectivity ARCs (1.476 ST) than for continuative ARCs (0.569 ST; KS: $D = 0.4667$, $p < 0.05$). However, Relevance ARCs show no significant difference with either continuative (0.569 ST; KS: $D = 0.3922$, $p \approx 0.09$) or subjectivity (1.476 ST; KS Test: $D = 0.1667$, $p \approx 0.309$) ARCs.

The results provided above describe onset as an absolute phenomenon (without taking the preceding IU into account). A more relative conception of onset values (onset normalised differential values) provides complementary information. In this perspective, it is interesting to note that significantly higher differential onset values were found for subjectivity ARCs (2.383 ST) as opposed to relevance ARCs (1.113 ST; KS: $D = 0.2425$, $p < 0.05$). Continuative ARCs also display such higher differential onset (2.088 ST; KS: $D = 0.498$, $p < 0.05$), but without any significant difference with subjectivity ARCs (KS: $D = 0.3444$, $p \approx 0.235$). Figure 4 illustrates these results.

5. Discussion

Our analyses have shown that ARCs display features that are both traditional and atypical compared with a general category of parentheticals if we consider

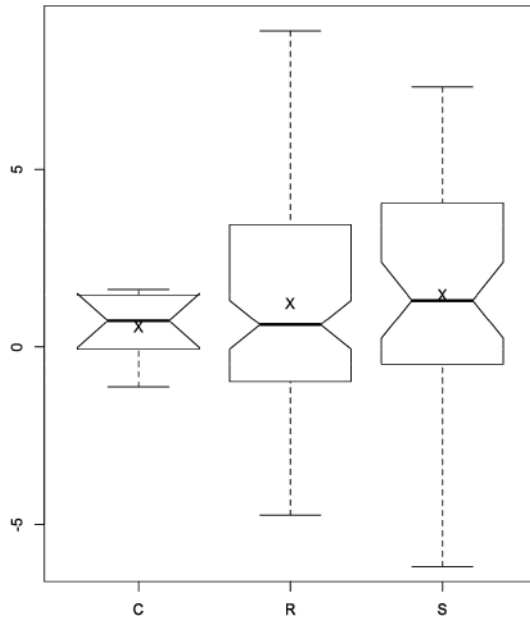


Figure 3. Normalised onset values by ARC type (*X* symbols mark sample means)

the literature on the subject. This demonstrates the descriptive inadequacy of a global characterisation of parenthetical structures, based on such traditionally advocated characteristics as lower pitch (Crystal 1969; Bolinger 1989; Local 1992), diminished loudness (Crystal 1969; Local 1992) and onset depression (Wichmann 2000). Table 6 schematically summarizes the differences between traditionally assumed characteristics and the results based on our study.

We have thus shown that, in comparison with surrounding IUs, ARCs display

- identical pitch spans, as opposed to the narrower spans associated with parentheticals,
- identical intensity levels and spans, as opposed to the lower levels and narrower spans observed with parentheticals,
- higher onset absolute and positive onset differential values, as opposed to the lower absolute and negative differential values attached to canonical parentheticals.

Our work is therefore in line with such studies as Kaltenböck (2007), who proposes a taxonomy of spoken parentheticals, Dehé and Wichmann (2010), who focus on comment clauses, or Dehé (2009), who discriminates between

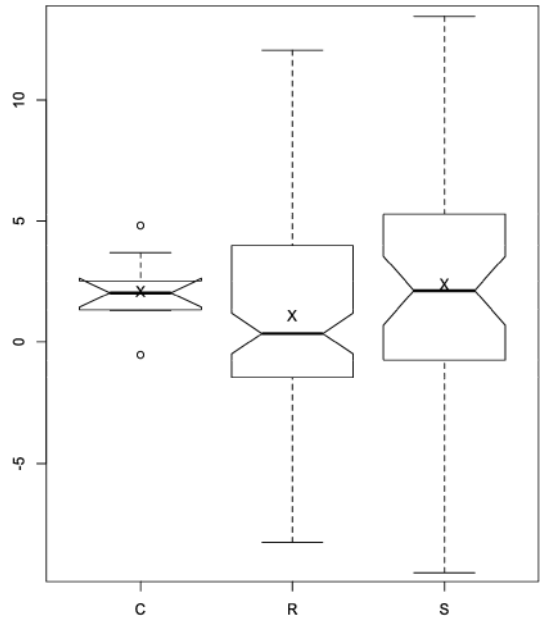


Figure 4. Normalised differential onset values by ARC type

Table 6. Global results for ARCs compared with canonical parentheticals (lower means “lower than both the preceding and the following IU”, etc.)

Prosodic feature		Canonical parentheticals	ARCs
Pitch	Level	Lower	Lower
	Span	Narrower	Identical
Intensity	Level	Lower	Identical
	Span	Narrower	Identical
Onset	Absolute	Lower	Higher
	Differential	Negative	Positive

three types of parentheticals: ARCs, sentential parentheticals and comment clauses. Specifically, our work indicates that such a taxonomy should not only be based on formal distinctions, but also on discourse considerations.

The complex picture drawn from our global results seems to benefit from the differential approach based on Loock’s categories. Indeed, we have shown that, taken globally, ARCs display typical cues of continuity (constant pitch span and intensity level and span) and discontinuity (positive onset values, lower pitch level); such ambivalent behaviour becomes clearer when the three cate-

gories of continuative, relevance and subjectivity ARCs are discriminated between. Our differential results provide evidence in favour of the existence of a (dis)continuity continuum along which ARC types can be arranged.

If the four prosodic parameters of onset value, onset differential, pitch level and pitch span are to be retained, continuative ARCs can unsurprisingly be regarded as the most discourse integrated clauses: out of the 4 parameters, they display 3 typical cues of continuity (no significantly positive onset values, no significant difference in pitch and intensity levels with surrounding IUs); discontinuity is cued only through significantly high onset differential values. However, our results concerning pitch level, may need more cautious exploitation. As can be seen from table 3, all 10 continuative ARCs in our corpus are sentence final. Due to declination, this potentially explains the negative values found (see table 5) for pitch level (−0.929 ST compared with −0.609 ST for relevance ARCs and −0.529 ST for subjectivity ARCs). It also explains the negative values found for the preceding IU (−0.259 ST compared with 0.447 ST for relevance ARCs and 0.099 ST for subjectivity ARCs), together with the positive values measured for the following (necessarily sentence initial) IU (0.128 ST compared with −0.045 ST for relevance ARCs and −0.157 ST for subjectivity ARCs).

At the other end of the continuum, subjectivity ARCs, characterised mostly by discontinuity cues, seem to be the least discourse integrated clauses: out of the 4 parameters, they display 3 typical cues of discontinuity (significantly positive onset values, higher onset differential values and higher pitch level), constant intensity level being the only cue of continuity.

In between these categories, relevance ARCs share both continuity and discontinuity cues. Continuity is marked by the absence of any significant onset differential values, and constant intensity level; discontinuity shows through significantly positive onset values and lower pitch level. All these elements are represented schematically in table 7 and figure 5.

The information conveyed by a subjectivity ARC clearly departs from the discourse topic or foreground, such as suggested by Hopper and Thompson (1980), who define foreground as “compris[ing] the backbone or skeleton of the text, forming its basic structure” as opposed to background information,

Table 7. *Continuity (C) and discontinuity (D) marking in relation to ARC type and prosodic parameter*

ARC type	Parameters Onset value	Onset differential	Pitch level	Intensity level
Continuative	C	D	C	C
Relevance	D	C	D	C
Subjectivity	D	D	D	C

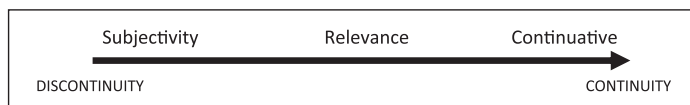


Figure 5. *(Dis)Continuity gradient of ARC types*

which “put[s] flesh on the skeleton, but [is] extraneous to its structural coherence” (Hopper and Thompson 1980: 281). The prosodic patterns observed thus seem to confirm Loock’s idea of a rupture between the referential and interpretative levels.

Relevance-oriented information, on the other hand, is more integrated with the discourse topic than subjective information. The purpose of relevance ARCs is neither narrative progression (as for continuative ARCs) nor departure from the discourse topic through subjective asides, but the optimisation of the relevance of the information conveyed by the main clause (current discourse topic). As such, relevance ARCs provide background information summoned to support foreground information in a closer relationship with the discourse topic than that observed with subjectivity ARCs, which induce a more important shift away.

Our study confirms that, in comparison with relevance and subjectivity ARCs, the status of continuative ARCs is exceptional: in spite of their being subordinate clauses, they introduce new extra-linguistic events and play an important part in narrative dynamism traditionally restricted to main clauses (see Depraetere 1996). This category of ARCs provides smooth narrative progression, prosodically marked by continuity cues in spite of the discourse/hearer new information introduced. It should however be remembered that the data set corresponding to continuative ARCs in our corpus is limited to 10 items, and would obviously require more exhaustive investigation.

6. Conclusion

Our study sheds some light on the complex behaviour adopted by ARCs as a whole, with both traditional and atypical characteristics as compared with a general category of parentheticals. A potential explanation for this behaviour could be that various discourse functions associated with one specific syntactic structure give way to differences in prosodic realisation. In this perspective, we have shown that a finer-grained approach to parentheticals and ARCs in particular provides new insight: more specifically, onset value, onset differential and pitch level vary according to the discourse function fulfilled by ARCs. We thus suggest the existence of a continuity gradient along which ARC types can be arranged, with subjectivity ARCs as the most discontinuous of the three

types, continuative ARCs representing the most integrated information, while relevance ARCs occupy an intermediate position, in line with Cornilescu (1981)'s concept of 'not so hot news'. It is important to note that the prosodic patterns observed in our data may be associated with a variety of discourse or conversational functions; for instance, Local (2005) observes similar prosodic characteristics with other, non-parenthetical structures, i.e. collaborative turn-completions. In line with previous work (e.g. Di Cristo et al. 2004), this underlines the variability and multi-functionality of prosodic features. Finally, our study also questions the unity of a broad category of parentheticals, with a wide spectrum of both prosodic configurations and discourse functions.

Our analyses raise a number of questions that obviously require further investigation. First, among the morpho-syntactic, semantic and pragmatic parameters used in the discourse annotation, this study focuses on the type and sentence position of ARCs; other parameters, in particular the nominal *vs.* sentential status of the antecedent and topical structure need to be taken into account. A potential follow up to this study may also regard the construction and evaluation of a predictive model capable of identifying the ARC type based on prosodic features; one possible method would rely on classification regression algorithms such as CART (Breiman et al. 1984), but may require a more important corpus, notably concerning continuative ARCs.

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Bionotes

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Notes

1. The preliminary results of a pilot study were presented at the workshop “Constraints in Discourse II” that was held in Dublin, Ireland, in July 2006 and were consequently published in the proceedings of this conference. Subsequent results were presented at the LAGB 2008 conference held in Colchester in September 2008. We would like to thank the organisers as well as the reviewers and the audience for their useful comments.
2. The existence of ARCs as a single structure is itself debatable, some authors (Canac Marquis and Tremblay 1996; De Vries 2006) claiming that they are nothing but determinative relative clauses (DRCs) in apposition. Even the syntactic unity of DRCs has been questioned (see Fox 1994; Fox and Thompson 2007).
3. The opposition between ARCs and DRCs has been given many different labels: restrictive *vs.* non-restrictive, determinative *vs.* appositive, determinative *vs.* descriptive, restrictive *vs.* appositive, defining *vs.* non-defining, attached *vs.* detached, integrated *vs.* supplementary. Because of the many syntactic, semantic and pragmatic similarities with appositives in general (see Loock 2005, 2010, or O’Connor 2008) we use the label ‘Appositive Relative Clauses’ in this study.
4. Original corpus-specific transcription conventions have been retained, hence the non-uniformity in the presentation of examples.
5. Though somewhat counterintuitive, examples of initial ARCs (*i.e.* preceding their antecedent) can be found in structures such as *He quit his job and, which was really surprising, found another one the next day.*
6. Kolmogorov-Smirnov tests (henceforth “KS tests”) were used throughout the analyses; contrary to ANOVAs or pairwise *t*-tests, these tests do not presuppose any homoscedasticity of the distributions.

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